

Chapter 1 — Abstract setting

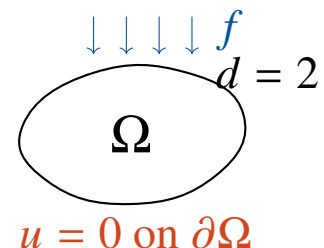
The multidimensional problem

MTH3340 · Numerical Methods for PDEs

1 The multidimensional problem

$$J(u) = \int_{\Omega} \frac{1}{2} |\nabla u|^2 - f u \, dx = \frac{1}{2} a(u, u) - \ell(u),$$

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v, \quad \ell(v) = \int_{\Omega} f v \quad \text{a QMP.}$$



Max. functional space = $\{v : a(v, v) < \infty\}$ with the bc's $\longrightarrow H_0^1(\Omega)$, a Hilbert space (previous lecture!), and $a(u, u) > 0 \ \forall u \in H_0^1(\Omega) \setminus \{0\}$ (PD).

Continuity, $|\ell(v)| \leq C \|v\|_{H_0^1}$? By Cauchy–Schwarz,

$$\ell(v) = \int_{\Omega} f v \leq \left(\int_{\Omega} f^2 \right)^{1/2} \left(\int_{\Omega} v^2 \right)^{1/2} = \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \stackrel{?}{\leq} C \|v\|_{H_0^1(\Omega)}.$$

$f \in L^2(\Omega) \longrightarrow \ell$ is continuous in $H_0^1(\Omega)$ — if we can bound $\|v\|_{L^2}$ by $\|v\|_{H_0^1}$ (next).

First Poincaré–Friedrichs inequality

Theorem (First Poincaré–Friedrichs inequality)

Given a bounded domain $\Omega \subset \mathbb{R}^d$, it holds

$$\|u\|_{L^2(\Omega)} \leq \text{diam}(\Omega) \|\nabla u\|_{L^2(\Omega)}, \quad \forall u \in H_0^1(\Omega).$$

Proof: lecture notes.

Define $\|u\|_{H_0^1(\Omega)} \doteq \left[\int_{\Omega} |\nabla u|^2 \right]^{1/2}$ ($= \|u\|_{H^1(\Omega)}$). The theorem states $\int_{\Omega} u^2 \leq C \int_{\Omega} |\nabla u|^2$, i.e., $\|u\|_{L^2(\Omega)} \leq C \|u\|_{H_0^1(\Omega)}$. Thus, if $u \in H_0^1(\Omega)$,

$$\|u\|_{H^1(\Omega)} = \left(\underbrace{\int_{\Omega} u^2}_{\|u\|_{L^2}^2} + \underbrace{\int_{\Omega} |\nabla u|^2}_{\|u\|_{H_0^1}^2} \right)^{1/2} \leq C \|u\|_{H_0^1(\Omega)}.$$

On $H_0^1(\Omega)$, $\|\cdot\|_{H_0^1}$ is a norm, equivalent to the full H^1 norm. And continuity follows: $\ell(v) \leq \|f\|_{L^2} \|v\|_{L^2} \leq \text{diam}(\Omega) \|f\|_{L^2} \|v\|_{H_0^1(\Omega)} \quad \checkmark$

Non-homogeneous bc's

Take now $u = g$ on $\partial\Omega$, and define the trial space

$$H_g^1(\Omega) \doteq \{v \in H^1(\Omega) : v = g \text{ on } \partial\Omega\}.$$

$$u = g \text{ on } \partial\Omega$$


Variational problem

$$u \in H_g^1(\Omega) : \int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} f v \quad \forall v \in H_0^1(\Omega).$$

Trial space $H_g^1(\Omega)$, test space $H_0^1(\Omega)$. Offset function method: $u = u_0 + \delta u$ for some $u_0 \in H_g^1(\Omega)$,

$$\delta u \in H_0^1(\Omega) : \int_{\Omega} \nabla \delta u \cdot \nabla v = \underbrace{\int_{\Omega} f v - \int_{\Omega} \nabla u_0 \cdot \nabla v}_{\tilde{\ell}(v)} \quad \forall v \in H_0^1(\Omega).$$

$$f \in L^2(\Omega), u_0 \in H^1(\Omega) \implies \tilde{\ell}(v) \leq \|f\|_{L^2} \|v\|_{L^2} + \|\nabla u_0\|_{L^2} \|\nabla v\|_{L^2} \leq C \|v\|_{H^1(\Omega)}.$$

Well-posedness; functions vs. functionals

$H_0^1(\Omega)$ is Hilbert and a is precisely its inner product (previous lecture!); ℓ is continuous (Poincaré) $\longrightarrow$ WP, $\exists! u$ of the problem (theorem of the previous lecture).

Note: f as a *functional* on $H_0^1(\Omega)$,

$$f : H_0^1(\Omega) \rightarrow \mathbb{R}, \quad v \mapsto f(v), \quad \text{vs. a function, } f(x), \quad x \in \Omega.$$

A function $h : \Omega \rightarrow \mathbb{R}$, $h(x)$, induces the functional

$$H(\cdot) \doteq \int_{\Omega} h(x) [\cdot] : H_0^1(\Omega) \rightarrow \mathbb{R}, \quad H(v) = \int_{\Omega} h(x)v(x).$$

But not every functional comes from a function, e.g., $H(w) \doteq w(0.5) \dots$

The dual space $H^{-1}(\Omega)$

Write the linear form as $\ell(v) = \int_{\Omega} f v \doteq f(v)$. Then

$$\boxed{f(v) \leq C \|v\|_{H_0^1(\Omega)}} \quad \longleftarrow \text{continuity} \quad \longrightarrow \quad f \in H^{-1}(\Omega), \quad \|f\|_{H^{-1}} < \infty,$$

the **dual space** of $H_0^1(\Omega)$, with the dual norm

$$\|f\|_{H^{-1}(\Omega)} \doteq \sup_{v \in H_0^1(\Omega) \setminus \{0\}} \frac{f(v)}{\|v\|_{H_0^1(\Omega)}}.$$

$f \in L^2(\Omega) \Rightarrow f \in H^{-1}(\Omega)$ (but not conversely!).

Ex.: $\Omega = [0, 1]$, point load at the midpoint,

$$\boxed{f(v) = v(1/2)} \quad (\text{a Dirac delta}):$$

$f \in H^{-1}(\Omega)$ but $f \notin L^2(\Omega)$ ($\int \delta^2 \neq \infty$).

